Quiz 1 Answer Key (Instructor Only): Foundations and Python Basics

Weeks 1–2 · 10 marks · Difficulty: low. Do not share this document with students.

Expected results. A prepared class should score 7 to 9. A class average below 6 means Week 1 or Week 2 needs a short review before moving on; check which question was missed most.

Section A: Multiple choice (1 mark each)

Q	Answer	Why it is right	Why the others are wrong
1	**B**	Data Science is the whole process: question, data, quality, analysis, communication, decision.	A is only one tool (machine learning). C is a skill, not the field. D is data engineering.
2	**A**	Data Engineers build the pipelines that collect, store and prepare data.	Statisticians analyse and infer; ML researchers develop learning methods; designers present.
3	**B**	One row is one order, so three orders are three rows. This is the grain question from Week 1.	A confuses rows with customers. C and D do not follow from the table.
4	**C**	Free text has no fixed fields or layout.	A and D are structured tables. B is semi-structured: tagged and nested but not a flat table.
5	**B**	A string and an integer are different types, so they are not equal.	A would be true only if Python converted automatically, which it does not. C: comparing never raises an error here. D is not a Boolean.
6	**C**	A dict holds named fields, so one CSV row is naturally a dict.	A and B are ordered by position, not by name. D has no order or names.
7	**B**	The start is included and the end is excluded: positions 1, 2 and 3 give 450, 3000 and 80.	A includes position 4. C starts at 0. D starts at position 2.

Q	Answer	Why it is right	Why the others are wrong
8	**B**	The session keeps the value in memory from the earlier run. Restart and Run all would reveal the problem.	A is not an explanation. C: nothing is saved to disk. D: notebooks run only the cells you run.

Section B: Short conceptual questions (1 mark each, half marks allowed)

Question 9: three bad rows with N/A.

Full mark (1): the program should keep running, record the rejected rows (and the reason), and report the count at the end, because stopping loses all the work for three rows and skipping silently hides a problem that changes the results. A very good answer also says the owner should decide whether to fix them.

Half mark (0.5): mentions only skipping, or only recording, or only catching the error, without saying why or without reporting the count.

Zero: the program crashes or the rows are silently dropped or replaced with 0 with no record.

Question 10: why ask what one row represents before combining tables.

Full mark (1): if the tables are at different grains (for example one row per order and one row per customer per month), combining them matches every row against several rows, so a value can be counted several times and totals become wrong. A concrete example is expected: a customer with 2 orders and 3 months becomes 6 rows and the amounts are counted three times.

Half mark (0.5): says that it helps you understand the data, or that rows may duplicate, without a consequence or example.

Zero: an answer unrelated to row meaning, such as “so the code runs faster”.

After the Quiz (5 minutes)

- Ask for a show of hands on each wrong answer. Question 3 and Question 10 together test whether the grain idea has landed, so spend a minute on them.
- If many chose A on Question 5, run `'24' == 24` and `type()` live.

- If many chose D on Question 8, repeat the hidden-state demo: define `x`, delete the cell, still use `x`, then Restart and Run all.